



S-SPARC

Specific Smart Prompting Assistant for perfoRmanCe

Presented by

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PROJECT VISION

Teaching Students to Think Before They Prompt

AI literacy and scaffolded prompting for higher education.

UNU Global Youth AI Future Innovation Competition 2026

Track: AI for Education · SDG 4: Quality Education

Phase 1 pilot in machine learning and programming courses; broader disciplines planned.

AI Is Everywhere in the Classroom: Teaching needs to change

The Vision

Building an AI collaborator that fosters active understanding rather than quietly replacing it.

S-SPARC is built on one premise: a tutoring AI should make a student work a little harder for the right reasons, not hand over answers the instant they're asked.

Challenges in AI-Driven Education

- Rapid, unstructured AI adoption is reshaping how students work
- Contextless prompts ("fix this script") let the AI do the reasoning
- Copy-paste outputs quietly erode critical thinking skills
- Without retrieval, repeated queries can trigger another model call

What Makes a Better Design Possible

- Reusing verified answers cuts cost without cutting learning value
- Structured prompting can itself be taught as a skill, not assumed
- Response depth can be matched to what a student is ready for

AI Tools Are Smart. Few Are Built to Teach.

Where Most EdTech AI Stops

- Usability and interface polish
- Ethics guidelines and disclosure
- Post-hoc plagiarism detection
- General-purpose tutoring chatbots

What It Rarely Does

- Enforce a minimum reasoning effort before answering
- Treat prompting itself as a teachable skill
- Stay affordable for under-resourced campuses
- Show students the cost of what they're asking for

What Our Own Data Showed

- Students defaulted to one-line, contextless prompts
- More AI usage did not predict better outcomes
- Prompt quality was positively associated with judged response quality

The problem isn't that students use AI. It's that nothing asks them to think first.

Introducing S-SPARC

Not just answering questions — shaping how students learn to ask them.



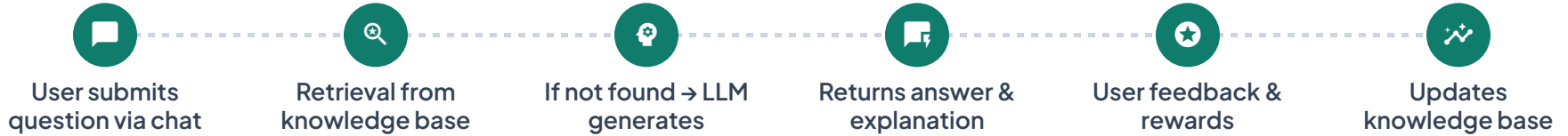
- Answers immediately, however the question is phrased
- No structural check on reasoning effort before responding
- Every question is re-computed, even those asked many times before
- No visibility into computational or environmental cost



- **Diagnostic gate:** requires a genuine, structured question first
- **Retrieval-first:** verified answers reused instantly, at no extra cost
- **Adaptive depth:** matched to the Bloom's-Taxonomy mode selected
- **3-tier hints:** delay full solutions until real struggle has happened

Not a faster way to get answers — a better way to learn how to find them.

How it Works



01

Student Query

Students submit general academic queries via the chat interface to begin processing (initial pilot testing conducted in IT/CS courses).



02

Diagnostic Check

C-I-O-E validation (Context, Input, Output, and Error Trace) & 200-character limit. Real-time spam filtering using Shannon entropy.



03

Semantic Cache

Search via FAISS. If the similarity is 0.90, the answer is returned instantly (<45 ms) without an external LLM.



04

LLM Fallback

A new query has been sent to the Cloud LLM. If it's offline, it automatically switches to the local Ollama server.

Grounding AI Tutoring in Constructivist Theory & Bloom's Taxonomy

Connecting cognitive development principles directly to operational AI interaction modes.

Theoretical Foundation



Active Knowledge Construction

01

(Piaget / Vygotsky): Learners build internal mental models actively. AI prompts reflection before revealing full solutions.



Zone of Proximal Development - ZPD

02

(Vygotsky): Targets space between unassisted capability and guided support, calibrating guidance to readiness.



Scaffolding & Gradual Release

03

(Bruner): Applies structured fading: high support early, systematically reducing assistance as mastery grows.

CORE PRINCIPLE: PRODUCTIVE FRICTION

Withholding instant answers is designed to preserve active critical thinking over copy-pasting, turning AI into a cognitive collaborator.

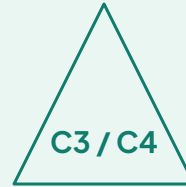


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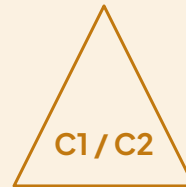
Bloom's Taxonomy Implementation



APPLY & ANALYZE

Targeted Analytical Mode

Targeted guidance and minimal delta fixes stripped of full solutions to isolate and solve logical errors.



REMEMBER & UNDERSTAND

Conceptual Validation Mode

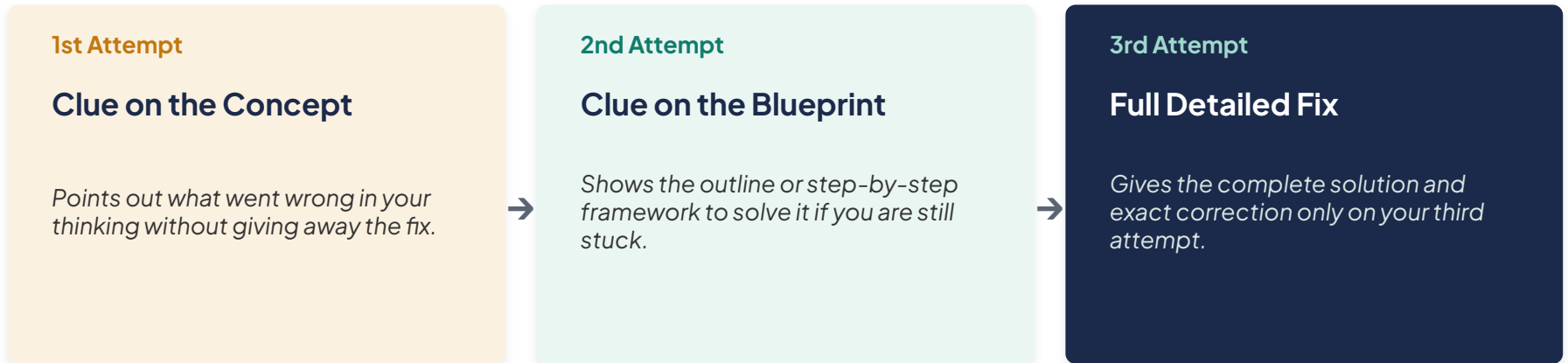
Analogies and core summaries without direct answers to force active student synthesis and mental mapping.

Note:

Intentionally aligned with Indonesian Undergraduate Learning Outcomes; Phase 1 focuses on Bloom's Taxonomy C1–C4, with advanced scaffolding reserved for later evaluation.

The 3-Tier Hint Progression

S-SPARC never gives full answers right away: hints unlock one step at a time as students keep trying.



Progressive hints preserve room for student reasoning before a detailed fix.

Keeping the Knowledge Base Honest

Every reused answer is pre-verified, so no student inherits someone else's mistake.

Five Checks Before an Answer Is Reused

1. **Accuracy** — Verifies technical and factual correctness of the answer.
2. **Logical correctness** — Validates sound step-by-step reasoning and coherence.
3. **Relevance** — Ensures strict alignment with student prompt & context.
4. **Readability** — Guarantees clear, structured, and accessible language.
5. **Content quality** — Evaluates instructional clarity and scaffolding.

Ongoing Safety Nets

Outlier Detection

Statistical outlier flagging on anything that looks off

Automated Hygiene

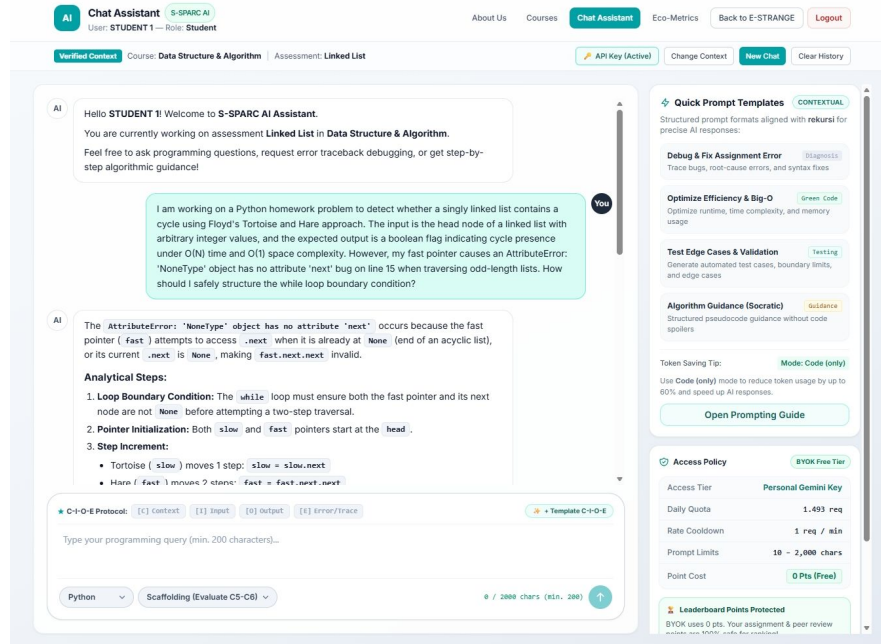
Duplicate and low-quality entries removed automatically

Data Resilience

Regular backups so nothing is lost by accident

S-SPARC Interface: Scaffolding Student Inquiry at Scale

Integrating Socratic guidance, structured prompts, and pedagogical rate-limiting.



AI Chat Assistant S-SPARC AI
User: STUDENT 1 — Role: Student

Verified Context Course: Data Structure & Algorithms Assessment: Linked List

API Key (Active) Change Context New Chat Clear History

AI
Hello STUDENT 1! Welcome to S-SPARC AI Assistant.
You are currently working on assessment **Linked List** in **Data Structure & Algorithms**.
Feel free to ask programming questions, request error traceback debugging, or get step-by-step algorithmic guidance!

You
I am working on a Python homework problem to detect whether a singly linked list contains a cycle using Floyd's Tortoise and Hare approach. The input is the head node of a linked list with arbitrary integer values, and the expected output is a boolean flag indicating cycle presence under O(N) time and O(1) space complexity. However, my fast pointer causes an `AttributeError: 'NoneType' object has no attribute 'next'` bug on line 15 when traversing odd-length lists. How should I safely structure the while loop boundary condition?

AI
The `AttributeError: 'NoneType' object has no attribute 'next'` occurs because the fast pointer (`fast`) attempts to access `.next` when it is already at `None` (end of an acyclic list), or its current `.next` is `None`, making `fast.next.next` invalid.

Analytical Steps:

- Loop Boundary Condition:** The `while` loop must ensure both the fast pointer and its next node are not `None` before attempting a two-step traversal.
- Pointer Initialization:** Both `slow` and `fast` pointers start at the `head`.
- Step Increment:**
 - Tortoise (`slow`) moves 1 step: `slow = slow.next`
 - Hare (`fast`) moves 2 steps: `fast = fast.next.next`

C-I-O-E Protocol: [C] Context [I] Input [O] Output [E] Error/Trace

Type your programming query (min. 200 characters)...

Python Scaffolding (Evaluate CS-C6)

2000 chars (Min. 200)

Quick Prompt Templates CONTEXTUAL
Structured prompt formats aligned with *rekursi* for precise AI responses:

- Debug & Fix Assignment Error** *Diagnosis*
Trace bugs, root-cause errors, and syntax fixes
- Optimize Efficiency & Big-O** *Green Code*
Optimize runtime, time complexity, and memory usage
- Test Edge Cases & Validation** *Testing*
Generate automated test cases, boundary limits, and edge cases
- Algorithm Guidance (Socratic)** *Guidance*
Structured pseudocode guidance without code spoilers

Token Saving Tip: Use Code (only) mode to reduce token usage by up to 60% and speed up AI responses.

Mode: Code (only)

Open Prompting Guide

Access Policy BYOK Free Tier

Access Tier	Personal Gemini Key
Daily Quota	1,493 req
Rate Cooldown	1 req / min
Prompt Limits	10 - 2,000 chars
Point Cost	0 Pts (Free)

Leaderboard Points Protected
BYOK uses 0 pts. Your assignment & peer review points are 100% safe & guaranteed.

Context-Aware Scaffolding

Delivers structured, step-by-step conceptual hints rather than direct code outputs to preserve problem-solving ownership.

Structured Prompt Templates

Guides students toward high-order inquiry (e.g., edge-case validation, algorithm optimization) instead of low-effort debugging requests.

Enforced Productive Friction

Built-in rate limits (1-min cooldown) and character thresholds (200–2,000 chars) prevent passive copy-pasting and encourage deep prompt formulation.

Curriculum Alignment

Binds AI context strictly to specific course modules and verified topics (e.g., Data Structures & Algorithms).

*Note: UI display reflects the initial IT pilot study; platform architecture is engineered for multi-disciplinary expansion.



Semantic Caching: Eliminating Redundant LLM Inferences

Bypassing live LLM inferences for repeated student queries using vector similarity.

The screenshot displays the S-SPARC AI Chat Assistant interface. At the top, it identifies the user as 'STUDENT 2' and the role as 'Student'. The interface includes navigation links like 'About Us', 'Courses', 'Chat Assistant', 'Eco-Metrics', 'Back to E-STRANGE', and 'Logout'. A 'Verified Content' badge is visible next to the course 'Data Structures & Algorithms' and the assessment 'Linked List'.

The chat history shows a previous interaction where the AI assistant welcomed the student and provided guidance on asking programming questions. The current query from the student is: "I am solving a Python assignment to detect if a singly linked list has a cycle using the Floyd's Tortoise and Hare approach. Given the head node of a linked list with integer values, the expected return is a boolean indicating cycle detection under $O(N)$ time and $O(1)$ space complexity. However, my fast pointer triggers an `AttributeError: 'NoneType' object has no attribute 'next'` error on line 15 when handling odd-length lists. How can I fix the while loop condition to prevent this bug?"

The AI assistant's response, labeled 'Zero-Token Vector Semantic Cache (Retrieved from DB)', explains that the error occurs because the fast pointer attempts to access `.next` when it is already `None` (end of an acyclic list), or its current `.next` is `None`, making `fast.next.next` invalid. It then provides 'Analytical Steps' to resolve the issue:

- Loop Boundary Condition:** The `while` loop must ensure both the fast pointer and its next node are not `None` before attempting a two-step traversal.
- Pointer Initialization:** Both `slow` and `fast` pointers start at the `head`.
- Step Increment:**

The interface also features a sidebar with 'Quick Prompt Templates' such as 'Debug & Fix Assignment Error', 'Optimize Efficiency & Big-O', 'Test Edge Cases & Validation', and 'Algorithm Guidance (Socratic)'. It includes an 'Access Policy' section showing 'BYOK Free Tier' details like 'Personal Gemini Key', 'Daily Quota' of 1,491 req, 'Rate Cooldown' of 1 req / min, 'Prompt Limits' of 10 - 2,000 chars, and 'Point Cost' of 0 Pts (Free). At the bottom, there's a 'Leaderboard Points Protected' section.

Zero-Token Retrieval

Detects semantically identical student queries and serves pre-validated answers directly from the vector database.

Sub-Second Latency & Zero Cost

Completely bypasses live LLM calls, reducing API inference costs to zero while delivering instant feedback.

Sustainable Infrastructure (Green AI)

Prevents redundant computations for high-frequency questions, minimizing the carbon and compute footprint of AI tutoring.

Verified Output Consistency

Ensures students receive pre-approved, pedagogically sound guidance for standard programming errors.

Experimental Design

Approach

- Quasi-experimental evaluation conducted in live courses rather than controlled lab simulations

Participants

- Undergraduate Computer Science students, cohorts 2021-2024
- Enrolled in a Machine Learning course at Maranatha Christian University

Setting

- Seven laboratory sessions across the semester
- Varying task difficulty and complexity per session



A real lab session from the multi-semester deployment.

What the Data Shows

TRL 6

Phase 1 CS lab proving ground
across live semesters

~84%

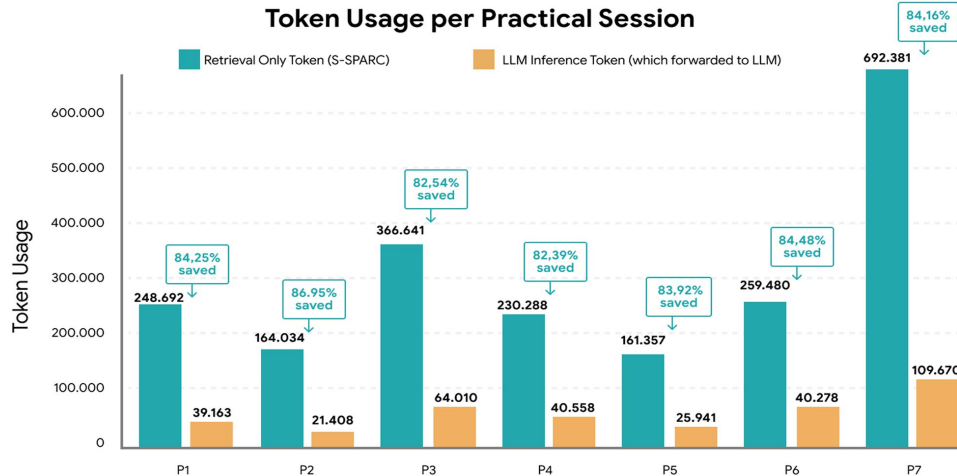
reduction in modeled
inference-token volume through
retrieval and local routing

~79%

more concise and on point prompt
sent to the AI model per query

7.4 → 1.8

average debugging exchanges per
problem, before vs. after



Phase 1 Proving Ground

- High-complexity CS lab serves as a Phase 1 Proving Ground prior to institutional rollout.
- Across all 7 sessions, most token volume was handled locally without external LLM inference.
- The seven-session pilot indicates stable operation under the tested workload.

Student Academic Performance Comparison

Control Group (2024/2025, N=52) vs. Intervention Group (2025/2026, N=68)

Key Takeaways

Individual Exams (Major Breakthrough)

The intervention proved most successful in boosting independent student mastery. Midterm scores saw a dramatic leap of nearly 20 points, while final exam results improved significantly ($p < .001$).

Daily Homework & Assignments

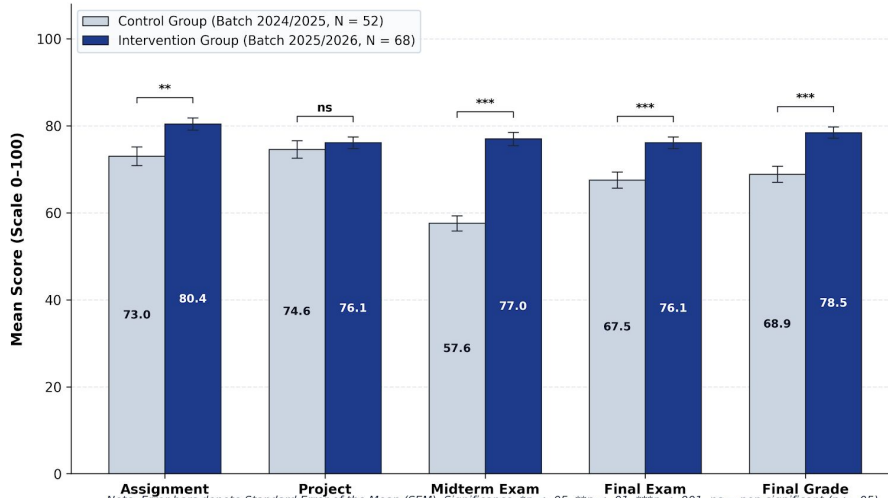
Learning discipline received a consistent positive push, showing statistically significant growth in assignment performance ($p = .004$).

Group Projects (Stable Baseline)

Team project scores remained stable and largely unaffected by the intervention, as baseline scores were already high ($p = .519$, non-significant).

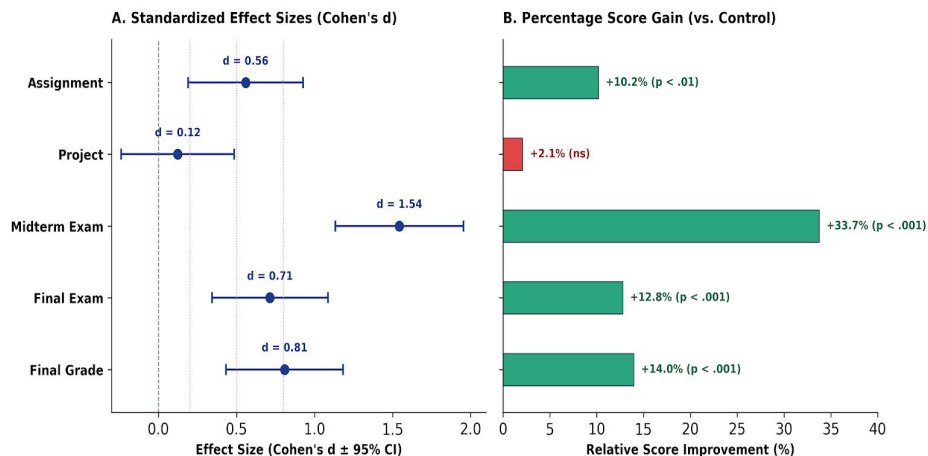
Overall Academic Grade Shift

Overall student standing shifted dramatically, elevating the cohort average from a C+/B- range to a solid B+/A- level ($p < .001$).



Intervention Efficacy & Pedagogical Insights

Understanding Effect Sizes and Key Takeaways for Instruction



Transformative Impact on Core Concepts

d = 1.54

Highest impact during the midterm phase, demonstrating a substantially stronger student grasp of foundational material.

Sustained Semester-Long Growth

d = 0.81

Early conceptual gains were retained through finals, culminating in a +14% total increase in course outcomes.

Steady Learning Habits

d = 0.56

Moderate improvements in daily homework (+10.2%) reflect a meaningful, positive shift in ongoing engagement.

Strategic Pedagogical Insight

d = 0.12

Minimal impact on group work indicates intervention specifically empowers individual cognitive processing.

Enhanced Prompting Literacy (Student Perception)

4.07 / 5.00

Post-course survey results confirm that using SSPARC significantly boosted student awareness of effective AI prompting across 27 evaluation metrics.

Consistency

System Reliability

- Recorded automated checks passed during the evaluation period; the repository test matrix provides the implementation detail.
- The evaluated retrieval threshold produced high precision on the current test set.
- The remaining gap was missed matches, which the next evaluation will address.

Early vs. Late Semester

- Modeled reduction stayed close from the first three weeks to the last three (about 86% in both periods).
- No detectable change was observed between early and late phases ($p = .273$).
- Local and retrieval paths remained active throughout the seven sessions.

The pattern remained similar across the seven-session pilot.

Future Work

Generalizability

- Every result so far comes from one institution, one LMS, and one set of CS courses
- Validated in a live multi-semester deployment rather than a lab simulation, though institutional generalizability remains to be proven.
- Phase 1 of the roadmap exists specifically to test a second course and a second LMS before any broader claim is made

Initial Friction, Then Adaptation

- Initial use created friction as students adapted to structured prompts and cooldowns.
- Post-course survey: 4.07/5 across 27 five-point Likert items ($n = 67$), indicating agreement that S-SPARC increased awareness of effective prompting.
- Independent learning and accessibility effects remain part of the next validation phase.

Alignment with the UN Sustainable Development Goals

SDG 4

Quality Education

Provides scalable tutoring in TA-scarce environments, enforcing active learning over passive answer copying. IT/CS serves as an initial high-complexity baseline, with multi-faculty expansion planned in upcoming roadmap phases.

SDG 9 & 10

Innovation & Reduced Inequalities

Roughly one in three questions is answered free, from a local cache, with no internet call; an offline fallback keeps students online even on a lean IT budget or unreliable connection.

SDG 17

Partnerships

Ensures vendor-neutral LMS integration (LTI 1.3 roadmap) without lock-in, enabling inter-university knowledge base sharing.

4 QUALITY
EDUCATION**9** INDUSTRY, INNOVATION
AND INFRASTRUCTURE**10** REDUCED
INEQUALITIES**17** PARTNERSHIPS
FOR THE GOALS

12–Month Roadmap

Each phase builds only on what the previous phase has earned the right to attempt.

Phase 1 · Months 0–3

Prove It Generalises

- Build the LTI 1.3 bridge for Moodle / Canvas
- Roll out to a second course or instructor at Maranatha
- Package the prompt builder as a self-contained module

Phase 2 · Months 3–6

First Outside Adopters

- Onboard 2–3 partner universities via the UNU network
- Run a paid pilot with at least one partner
- Add multilingual support and more domain knowledge & subjects

Phase 3 · Months 6–12

Scale on Evidence

- Controlled study across partner campuses
- Build the instructor analytics dashboard
- Set final pricing from real pilot conversations

Free Core, Paid Depth

Institution-Funded B2B Model: Zero Barrier for Students, High ROI for Universities

Free — Community

\$0 (Free Forever)

- Core Metacognitive Tutor & 60s cooldown
- Offline Fallback (Ollama engine)
- LTI 1.3 Basic integration

Target: Developing campuses & limited infrastructure (SDG 4 & 10)

Professional

Subscription (~\$1.50/student/year)

- Cloud Vector Semantic Cache
- Automated Verification Engine
- Student Eco-Metrics Dashboard

Target: Departments (LLM API Cost Efficiency)

Enterprise

Institutional Annual License

- Multi-Department Knowledge Sync
- Instructor Analytics & Pedagogical Insights
- Custom On-Premise & Dedicated SLA
- Expansion to non-IT faculties (STEM, Social Sciences, General Education)

Target: Large-scale / multi-campus universities

Financial ROI for Universities (Modeled Cost Reduction)

Modeled analysis indicates **up to ~84% lower inference-token demand** through semantic retrieval and local routing in the pilot workload.

Target Market & Go-To-Market Strategy

Initial Department Launch: Launches initially in high-volume CS/IT departments (~1,500 programs in SEA) to validate willingness-to-pay and drive immediate API cost reduction.

Campus-Wide Enterprise Scaling: Expands into full university-wide enterprise licenses across non-IT faculties (STEM, Social Sciences, and General Education).



Two Risks We Take Seriously

Offline AI May Underperform Premium Cloud Models

- Risk: the free offline fallback can write lower-quality code than a top-tier cloud model
- Mitigation: prompts are trimmed to essential information before sending
- Mitigation: quality screening is applied before generated answers enter the reusable knowledge base

Student Privacy & University Data Protection

- Risk: sensitive student data and usage patterns need real protection
- Mitigation: student data handling follows institution-defined controls; no student work is used to train outside models
- Mitigation: universities requiring local data governance can run the system on their own servers

Thank You

"The problem is not the AI, it's how we use it." S-SPARC changes that.



Who We Are

We are an AI innovation team from Maranatha Christian University, passionate about transforming education through artificial intelligence.

Key Initiatives & Expertise:

- Bringing together students & faculty across AI, software engineering, and education.
- Designing ethical, sustainable, and high-impact learning solutions.
- Empowering future-ready pedagogical tools for higher education.

Our Main Research Themes

- Academic integrity
- Educational computing
- Artificial intelligence

Contact & Collaboration

Contact us for inquiries or collaboration!

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S-SPARC Repository
(Scan QR Code)

